FINANCIAL MODELLING

Pro Forma Financial Statements for Colgate-Palmolive Company, 2024

Question: Pro forma income statement, balance sheet and cash flow statement

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Executive Summary

The pro forma statements built in this report give a coherent base case for Colgate-Palmolive Company in 2024: net sales of $19,184 million, EBITDA of $4,335 million, net profit of $2,688 million and total assets of $17,800 million. The income statement, balance sheet and cash flow statement tie to one another in every forecast year, and the 2022 starting balance sheet agrees to the reported total assets of $15,731 million.

The forecast is driven by one assumption more than any other. Over 2018 to 2022 Colgate's operating costs rose almost one for one with sales, and the regression used to project costs has a slope of 0.94. Each extra dollar of sales therefore adds only about six cents of EBITDA, so net profit stays close to $2.7 billion every year even though sales grow by 3.33% annually. A change of one percentage point in sales growth moves 2024 net profit by about $15 million. A change of four cents in the cost slope moves it by about $580 million. Readers should treat the cost slope as the assumption that needs the most scrutiny.

The base case is also conservative on sales. Colgate reported 2023 net sales of $19,457 million, 8.3% above 2022, while the model assumes 3.33% growth. The forecast of $18,565 million for 2023 is therefore 4.6% below the reported figure. Section 'Comparison with Reported 2023 Results and Limitations' sets out this gap and the other simplifications in the model.

The method follows the percent-of-sales approach taught in the Building Financial Statements lecture. Sales are projected from a growth rate, costs from a regression on sales, working capital and fixed assets from their historical ratios to sales, and cash is the balancing item. All calculations are in the accompanying Excel workbook, and every assumption is documented

there with its source.

Introduction

Purpose and scope

This report answers Question 1 of the BBA2293 Financial Modelling group assignment: prepare pro forma financial statements for Colgate-Palmolive Company for 2024, comprising an income statement, a balance sheet and a cash flow statement, using the method discussed in the lecture on Building Financial Statements. Questions 2 to 4 (free cash flow, the weighted average cost of capital and the share valuation) are covered by other members of the group and are outside the scope of this report.

The report has three aims: to explain how each forecast line is derived, to justify each assumption with evidence from the 2018 to 2022 financial statements, and to test how far the 2024 results depend on those assumptions. A comparison with Colgate's reported 2023 results is included to show how the forecast performs against actual outcomes.

Company background

Colgate-Palmolive is a global consumer products company. Its products fall into oral care, personal care, home care and pet nutrition. It reports in US dollars, and its sales are made in many countries, so its results are affected by currency movements as well as by volumes and prices. In 2022 it reported net sales of $17,967 million and net income attributable to the company of $1,785 million.

Data sources and modelling approach

The model uses five years of annual statements, 2018 to 2022, taken from the investor relations website of Colgate-Palmolive. The assignment specifies five years of statements and the 2023 statements were due only in the first quarter of 2024, so 2022 is the base year and 2023 is the first forecast year. 2024 is the target year for this question, and the model continues to 2027 so that the trend in each line is visible.

The model follows a sequence. First, the historical statements are reduced to a small set of ratios. Second, those ratios become assumptions, one per cell, each linked to the analysis that supports it. Third, the income statement is projected, then the balance sheet, with cash as the balancing item. Fourth, a cash flow statement is derived from the changes in the other two statements and checked against the balance sheet cash.

All figures in this report are in US dollars millions unless stated. Forecast figures are taken directly from the Excel workbook and are rounded to the nearest million.

Historical Performance, 2018 to 2022

Income statement

Table 1: Selected historical income statement data ($ million)

	2018	2019	2020	2021	2022
Net sales	**15,544**	**15,693**	**16,471**	**17,421**	**17,967**
Gross profit	9,231	9,325	10,017	10,375	10,248
Gross margin	59.4%	59.4%	60.8%	59.6%	57.0%
SG&A expenses	5,389	5,575	6,019	6,407	6,565
SG&A as % of sales	34.7%	35.5%	36.5%	36.8%	36.5%
Impairment charges	0	0	0	571	721
Operating profit	**3,694**	**3,554**	**3,885**	**3,332**	**2,893**
Operating margin	23.8%	22.6%	23.6%	19.1%	16.1%
Income before income taxes	3,464	3,301	3,647	3,087	2,660
Effective tax rate	26.2%	23.4%	21.6%	24.3%	26.1%
Net income incl. noncontrolling	**2,558**	**2,527**	**2,860**	**2,338**	**1,967**

	2018	2019	2020	2021	2022
interests					

Source: Colgate-Palmolive Company annual financial statements, as entered in the Income Statement sheet of the workbook.

Net sales grew from $15,544 million in 2018 to $17,967 million in 2022. Growth was slow in 2019 (1.0%) and faster in 2020 and 2021 (5.0% and 5.8%), then eased to 3.1% in 2022. Measured from 2017 net sales of $15,454 million, the five-year compound annual growth rate is 3.06%.

Profitability weakened toward the end of the period. Gross margin was 59.4% in 2018 and peaked at 60.8% in 2020, then fell to 57.0% in 2022, because cost of sales rose 9.6% in 2022 against sales growth of 3.1%. SG&A rose from 34.7% of sales in 2018 to 36.5% in 2022. Two impairment charges, $571 million in 2021 and $721 million in 2022, reduced operating profit, and operating margin fell from 23.8% in 2018 to 16.1% in 2022. Net income including noncontrolling interests fell 15.9% in 2022 to $1,967 million.

The effective tax rate varied between 21.6% (2020) and 26.2% (2018). It was 26.1% in 2022.

Balance sheet

Table 2: Selected historical balance sheet data ($ million)

	2018	2019	2020	2021	2022
Cash and cash equivalents	726	883	888	832	775
Receivables, inventories and other current assets	3,067	3,296	3,450	3,565	4,338
Property, plant and equipment,	3,881	3,750	3,716	3,730	4,307

	2018	2019	2020	2021	2022
net					
Goodwill and other intangibles	4,167	6,175	6,718	5,746	5,272
Total assets	**12,161**	**15,034**	**15,920**	**15,040**	**15,731**
Accounts payable and accrued income taxes	1,633	1,607	1,796	1,915	1,868
Long-term debt	6,354	7,333	7,334	7,194	8,741
Total shareholders' equity (Colgate-Palmolive)	(102)	117	743	609	401

Source: Balance Sheet sheet of the workbook.

Total assets rose from $12,161 million in 2018 to $15,731 million in 2022. Most of the increase came in 2019, when payments for acquisitions of $1,711 million lifted goodwill from $2,530 million to $3,508 million and other intangibles from $1,637 million to $2,667 million. Property, plant and equipment was stable at about $3.7 to $3.9 billion until 2022, when it rose to $4,307 million, in line with higher capital expenditure of $696 million.

Shareholders' equity is small, between negative $102 million and $743 million, because the company has repurchased large amounts of its own shares. Treasury stock stood at $25,128 million in 2022. This matters for the forecast: the model needs an explicit view on whether buybacks continue (see the assumptions section).

Working capital moved with sales. Receivables, inventories and other current assets rose from $3,067 million to $4,338 million, with inventories growing from $1,250 million to $2,074 million. Accounts payable and accrued income taxes rose more slowly, from $1,633 million to

$1,868 million.

Cash flow statement

Table 3: Selected historical cash flow data ($ million)

	2018	2019	2020	2021	2022
Net cash provided by operations	3,056	3,133	3,719	3,325	2,556
Capital expenditures	(436)	(335)	(410)	(567)	(696)
Dividends paid	(1,591)	(1,614)	(1,654)	(1,679)	(1,691)
Purchases of treasury shares	(1,238)	(1,202)	(1,476)	(1,320)	(1,308)
Proceeds from exercise of stock options	329	498	874	424	418
Net change in cash	(809)	157	5	(56)	(57)

Source: Cash Flow (Format 2) sheet of the workbook.

Operating cash flow was between $2,556 million and $3,719 million, and it fell to its lowest level in 2022 as inventories and receivables absorbed cash. Capital expenditure was between $335 million and $696 million. Dividends paid rose every year, from $1,591 million to $1,691 million. Share repurchases of $1.2 to $1.5 billion a year were partly offset by proceeds from stock option exercises, so net repurchases averaged about $0.8 billion a year. Year-end cash stayed between $726 million and $888 million, which shows that the company distributes most of its operating cash flow rather than accumulating it.

Forecast Assumptions and Justification

Every assumption sits in its own labelled cell on the Pro-forma sheet and is linked to a supporting analysis sheet. Table 4 summarises them, and the sections below explain the evidence for each.

Table 4: Summary of forecast assumptions

Item	Value used	Basis
Sales growth	3.33% a year	Average of industry forecast (3.6%) and historical CAGR (3.06%)
Operating costs excluding depreciation	-$3,174m + 0.9395 x sales	Regression of 2018 to 2022 costs on sales (R-squared 0.97)
Depreciation	Opening gross fixed assets / 17	Average ratio of gross PP&E to D&A, 2018 to 2022
Interest rate on borrowings	2.44%	Average of historical rate (2.08%) and effective rate in December 2023 (2.79%)
Interest base	$8,766m, constant	Notes payable, current portion of long-term debt and long-term debt at 2022
Interest earned on cash	1.0% of opening cash	Judgement, below the 3.88% 10-year Treasury yield
Current operating assets	21.26% of sales	Five-year average
Current operating liabilities	10.61% of sales	Five-year average
Net fixed assets	23.36% of sales	Five-year average
Tax rate	24.19%	Weighted average effective rate, 2018 to 2022
Dividend growth	1.54% a year	Compound growth of dividends paid, 2018 to 2022
Goodwill, intangibles and other assets	Constant at 2022	No acquisitions assumed

Item	Value used	Basis
Debt and other equity	Constant at 2022	No net borrowing, buybacks or share issues assumed

Sales growth

Sales grow at 3.33% a year. This is the average of two estimates. The first is the industry forecast of 3.6% a year for Colgate's net sales, recorded on the Sales Growth Assumption sheet. The second is the historical compound annual growth rate of 3.06%. Averaging an external forecast with the company's own record guards against relying on either alone. The year-by-year growth rates were uneven (1.0% to 5.8%), so a single average rate is a reasonable summary of a period with no clear trend.

Operating costs

Operating costs excluding depreciation comprise cost of sales, SG&A, other (income) expense and non-service postretirement costs, less depreciation and amortization. Interest is excluded because it is forecast separately. The goodwill impairments of 2021 and 2022 are also excluded: they are non-cash, and a model that repeated them every year would understate earnings.

These costs are projected from sales with a linear regression over the five historical years: costs = -3,174 + 0.9395 x sales. The fit is close (R-squared of 0.97). The slope is high because costs and sales moved together over the period. From 2018 to 2022 sales rose by $2,423 million and costs excluding depreciation rose by $2,462 million. The regression therefore assumes that Colgate cannot increase its margin by growing sales. This is a cautious assumption and it is the main driver of the forecast, as the sensitivity analysis shows.

The regression gives a falling EBITDA margin in the forecast, from 22.7% in 2022 to 22.6% in 2024 and 21.1% in 2027. The intercept of negative $3,174 million has no economic meaning on its own. It is what a straight line through five points with a slope near one requires, and it limits how far the forecast can be extended.

Depreciation

Depreciation each year equals the opening gross fixed assets divided by 17. The 17 comes from the ratio of gross property, plant and equipment to depreciation and amortization, which was between 16.0 and 17.6 years over 2018 to 2022 and averages 17 when rounded. D&A in the cash flow statement includes amortization of intangibles, so the ratio is an approximation. It is nevertheless the closest estimate of an average asset life available from the published statements.

Interest

Interest expense is the interest rate multiplied by interest-bearing borrowings of $8,766 million (notes payable, the current portion of long-term debt and long-term debt at the end of 2022). The rate is 2.44%, the average of two estimates. The first is the five-year average of net interest divided by borrowings, 2.08%. The second is the effective interest rate on Colgate's debt of 2.79% as of December 2023, taken from public data and recorded on the Sales Growth Assumption sheet. Using the same base for the rate and for the forecast keeps the interest charge consistent: forecast interest expense is $214 million a year, against net interest of $153 million in 2022.

Interest earned on cash is 1.0% of the opening cash balance. This is a judgement. Cash balances are small relative to sales, and the 10-year US Treasury yield at 30 December 2022 was 3.88%, so a lower return is assumed for operating cash. Using the opening balance rather than the average avoids a circular reference, because cash is the balancing item of the balance sheet.

Working capital and fixed assets

Current operating assets (receivables, inventories and other current assets) are projected at 21.26% of sales and current operating liabilities (accounts payable and accrued income taxes) at 10.61% of sales. Both are five-year averages. The ratio of current operating assets to sales ranged between 19.7% and 24.1%, and the ratio of current operating liabilities to sales between 10.2% and 11.0%, so the liability ratio is the more stable of the two. Cash is excluded from current assets, as the model treats it as the balancing item.

A consequence of using the five-year average is that net working capital falls from 13.7% of 2022 sales to 10.6% in 2023. This releases $391 million of cash in 2023 and is the reason why 2023 operating cash flow is higher than in later years. It assumes that the 2022 build-up in inventories and receivables reverses.

Net fixed assets are projected at 23.36% of sales, the five-year average of a ratio that stayed between 21.4% and 25.0%. Gross fixed assets are net fixed assets plus accumulated depreciation, and capital expenditure is the increase in gross fixed assets.

Tax, dividends and other items

The tax rate is 24.19%, the weighted average of the effective rate (total tax divided by total pre-tax income, 2018 to 2022). A weighted average is used instead of the simple average of 24.30% so that years with lower pre-tax income do not carry the same weight as years with higher income.

Dividends grow at 1.54% a year, the compound growth rate of dividends paid from $1,591 million in 2018 to $1,691 million in 2022. Dividend growth has been steady, but it has been slower than sales growth, and the model preserves that gap.

Goodwill, intangible assets, deferred tax assets and other assets are held at their 2022 values because the model does not forecast acquisitions. Debt is also held constant, and equity other than retained earnings is held constant, so the model assumes no net borrowing, no share buybacks and no share issues. These simplifications are discussed in the limitations section, because they affect the amount of cash that accumulates.

Pro Forma Income Statement

Table 5 shows the projected income statement. The 2024 column, the target year for this question, is shaded. The 2022 column is the base year, restated to exclude the impairment charge as explained below the table.

Table 5: Pro forma income statement ($ million)

	2022	2023	2024	2025	2026	2027
Sales	**17,967**	**18,565**	**19,184**	**19,822**	**20,482**	**21,164**
Operating costs excluding depreciation	(13,888)	(14,267)	(14,848)	(15,448)	(16,068)	(16,709)
EBITDA	**4,079**	**4,298**	**4,335**	**4,374**	**4,414**	**4,455**
EBITDA margin	22.7%	23.2%	22.6%	22.1%	21.6%	21.1%
Interest expense	(153)	(214)	(214)	(214)	(214)	(214)
Interest earned on cash	-	8	22	29	36	43
Depreciation	(545)	(564)	(599)	(642)	(689)	(739)
Profit before tax	**3,381**	**3,528**	**3,545**	**3,547**	**3,548**	**3,546**
Taxes	(693)	(854)	(858)	(858)	(858)	(858)
Profit after tax	**2,688**	**2,675**	**2,688**	**2,689**	**2,690**	**2,688**
Net profit margin	15.0%	14.4%	14.0%	13.6%	13.1%	12.7%
Dividends	(1,691)	(1,717)	(1,743)	(1,770)	(1,797)	(1,825)
Retained earnings for the year	997	958	944	919	892	863

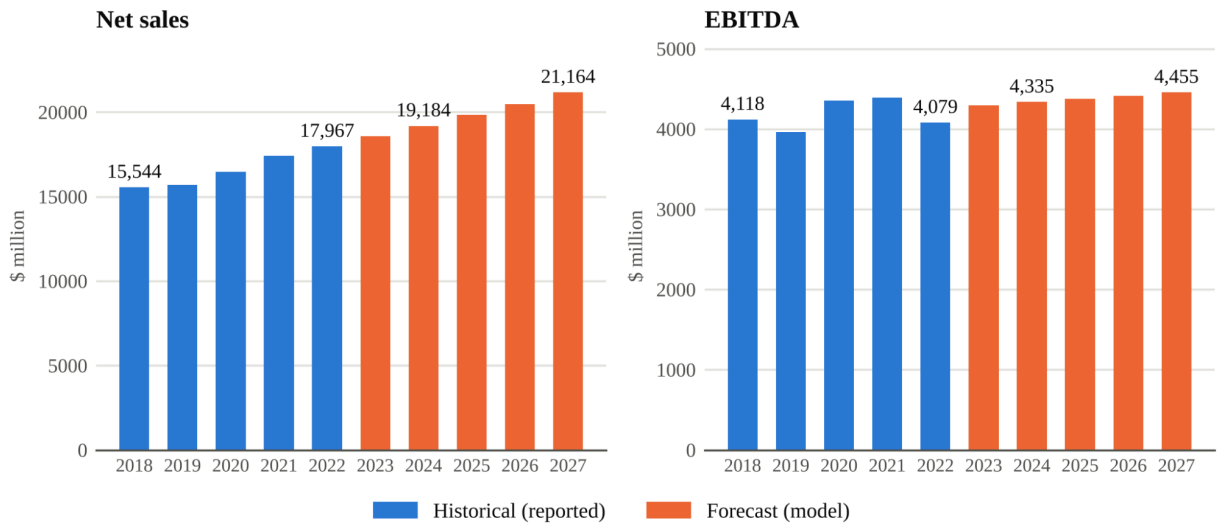
Source: Pro-forma sheet of the workbook, rows 21 to 33.

In 2024 the model projects sales of $19,184 million, which is 3.33% above the 2023 forecast of

$18,565 million. Operating costs excluding depreciation are $14,848 million, giving EBITDA of $4,335 million and a margin of 22.6%. Depreciation of $599 million, interest expense of $214 million and interest income of $22 million give profit before tax of $3,545 million. Tax at 24.19% is $858 million, leaving profit after tax of $2,688 million, or 14.0% of sales. After dividends of $1,743 million, $944 million is retained.

Figure 1 shows net sales and EBITDA from 2018 to 2027. Sales rise in every forecast year, from $17,967 million in 2022 to $21,164 million in 2027. EBITDA, by contrast, moves only from $4,079 million in 2022 to $4,455 million in 2027, and in 2024 it is close to the level reported in 2020 and 2021.

Figure 1: Net sales and EBITDA, 2018 to 2027 ($ million; blue = historical, orange = forecast)



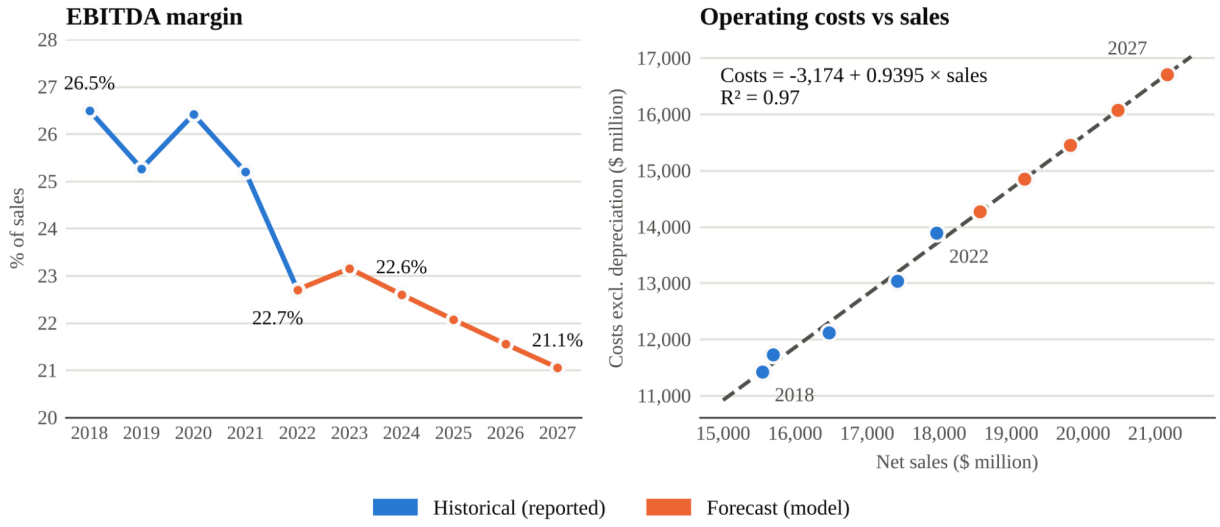
Source: Income Statement and Operating Cost Analysis sheets (2018 to 2022) and Pro-forma sheet (2023 to 2027). EBITDA is sales less operating costs excluding depreciation and excludes goodwill impairment charges.

Profit after tax is almost unchanged across the forecast, between $2,675 million and $2,690 million. This follows from the cost regression. Sales rise by about $620 to $680 million a year,

but costs rise by 94% of that amount, so EBITDA grows by only $37 to $41 million a year from 2024. Depreciation grows by $35 to $50 million a year with the asset base and absorbs most of that increase. The net profit margin therefore declines from 14.4% in 2023 to 12.7% in 2027.

Figure 2 shows the reason. On the left, the EBITDA margin falls from 26.5% in 2018 to 22.7% in 2022 and to 21.1% in 2027. On the right, the five historical observations lie close to the regression line (R-squared of 0.97) and the forecast points continue along the same line, so each additional dollar of sales carries about 94 cents of operating cost.

Figure 2: EBITDA margin and the operating cost regression



Source: Operating Cost Analysis sheet (2018 to 2022) and Pro-forma sheet (2023 to 2027). Historical points are blue and forecast points are orange; the dashed line is the regression of costs on sales.

The 2022 column is not the reported result. Reported 2022 net income including noncontrolling interests was $1,967 million. The base year in the model excludes the $721 million goodwill impairment, because the regression excludes impairment charges, and the tax charge is left at the reported $693 million. Adding $721 million to the reported figure gives the $2,688 million

shown. The base-year column is used only as the starting point for the growth calculations and is not a forecast.

Pro Forma Balance Sheet

Table 6: Pro forma balance sheet ($ million)

	2022	2023	2024	2025	2026	2027
Cash and marketable securities	775	2,196	2,930	3,631	4,299	4,931
Current operating assets	4,338	3,947	4,078	4,214	4,354	4,499
Fixed assets at cost	9,583	10,177	10,920	11,711	12,555	13,452
Accumulated depreciation	(5,276)	(5,840)	(6,438)	(7,081)	(7,770)	(8,508)
Net fixed assets	4,307	4,337	4,482	4,631	4,785	4,944
Goodwill, intangibles and deferred tax assets	5,407	5,407	5,407	5,407	5,407	5,407
Other assets	904	904	904	904	904	904
Total assets	**15,731**	**16,790**	**17,800**	**18,787**	**19,749**	**20,685**
Current operating liabilities	1,868	1,969	2,035	2,103	2,173	2,245
Debt and other financing	13,462	13,462	13,462	13,462	13,462	13,462

	2022	2023	2024	2025	2026	2027
liabilities						
Equity excluding retained earnings	(20,117)	(20,117)	(20,117)	(20,117)	(20,117)	(20,117)
Retained earnings and other comprehensive income	20,518	21,476	22,420	23,339	24,232	25,095
Total liabilities and equity	**15,731**	**16,790**	**17,800**	**18,787**	**19,749**	**20,685**

Source: Pro-forma sheet of the workbook, rows 37 to 51. Fixed assets at cost and accumulated depreciation are shown as the build-up of net fixed assets and are counted once in total assets.

Total assets in 2024 are $17,800 million, up from $15,731 million at the end of 2022. Net fixed assets are $4,482 million (23.36% of sales), supported by $10,920 million of gross assets and $6,438 million of accumulated depreciation. Current operating assets are $4,078 million and current operating liabilities $2,035 million, so net working capital is $2,043 million.

Cash is the balancing item. It is calculated as total liabilities and equity less all other assets, and it rises from $775 million at the end of 2022 to $2,930 million in 2024 and $4,931 million in 2027. The rise follows from the assumption that debt and share capital do not change: all profit not paid as dividends or invested in working capital and fixed assets stays in cash.

The financing side uses the operating approach from the lecture. Debt of $13,462 million is the total of the interest-bearing and other non-operating liabilities on the 2022 balance sheet: notes payable, the current portion of long-term debt, long-term debt, other accruals, deferred income taxes, other liabilities and noncontrolling interests. Operating liabilities are limited to accounts payable and accrued income taxes. Equity excluding retained earnings is negative $20,117

million, mostly the treasury stock held at cost. Retained earnings include accumulated other comprehensive loss and grow each year by profit after tax less dividends.

The 2022 column reconciles to the reported balance sheet. Total assets are $15,731 million and total liabilities and equity are $15,731 million, equal to the published figure. To achieve this, goodwill and other intangibles of $5,272 million were combined with the deferred tax asset of $135 million, and the unearned compensation of negative $1 million was included in equity.

Pro Forma Cash Flow Statement

The cash flow statement uses the indirect method. It starts from profit after tax, adds back depreciation, and adjusts for changes in working capital, then deducts investment in fixed assets and distributions to shareholders. Because debt and share capital are constant, financing consists only of dividends.

Table 7: Pro forma cash flow statement ($ million)

	2022	2023	2024	2025	2026	2027
Profit after tax	-	2,675	2,688	2,689	2,690	2,688
Add back depreciation	-	564	599	642	689	739
(Increase) decrease in current operating assets	-	391	(131)	(136)	(140)	(145)
Increase in current operating liabilities	-	101	66	68	70	72
Net cash from operating activities	**-**	**3,731**	**3,220**	**3,264**	**3,308**	**3,354**

	2022	2023	2024	2025	2026	2027
Capital expenditure	-	(594)	(743)	(792)	(843)	(898)
Change in goodwill, intangibles and other assets	-	0	0	0	0	0
Net cash used in investing activities	**-**	**(594)**	**(743)**	**(792)**	**(843)**	**(898)**
Dividends paid	-	(1,717)	(1,743)	(1,770)	(1,797)	(1,825)
Net borrowing	-	0	0	0	0	0
Net cash used in financing activities	**-**	**(1,717)**	**(1,743)**	**(1,770)**	**(1,797)**	**(1,825)**
Net increase in cash	**-**	**1,421**	**734**	**702**	**668**	**631**
Cash at beginning of year	-	775	2,196	2,930	3,631	4,299
Cash at end of year	**-**	**2,196**	**2,930**	**3,631**	**4,299**	**4,931**

Source: Pro-forma Cash Flow sheet of the workbook.

In 2024 operating activities generate USD 3,220 million: profit after tax of USD 2,688 million, depreciation of USD 599 million, an investment of USD 131 million in current assets, and an increase of USD 66 million in current liabilities. Capital expenditure is USD 743 million, which is the increase in gross fixed assets, and dividends are USD 1,743 million. Cash therefore rises by USD 734 million, from USD 2,196 million to USD 2,930 million.

Operating cash flow in 2023 is higher, at USD 3,731 million, because the move to average working capital ratios releases USD 391 million. The forecast of USD 3,220 million for 2024 is in the range of Colgate's reported operating cash flow in 2018 to 2021 (USD 3.1 to 3.7 billion) and above 2022 (USD 2,556 million). Capital expenditure of USD 743 million in 2024 is similar to the USD 696 million spent in 2022. It increases each year because gross fixed assets are projected to grow with sales while depreciation reduces the net balance.

Model Integrity Checks

Four checks confirm that the statements are consistent. They are built into the workbook so that they update if an assumption changes.

Check	Test	Result
Balance sheet balances	Total assets less total liabilities and equity	Zero in 2022 and in each forecast year
Cash flow ties to balance sheet	Cash at end of year (cash flow statement) less cash on balance sheet	Zero in each forecast year
Base year agrees to reported accounts	2022 total assets	USD 15,731 million, equal to the published balance sheet
Base-year profit reconciles	Reported 2022 net income plus impairment	USD 1,967m + USD 721m = USD 2,688m, as shown in the model

Sensitivity of the 2024 Forecast

Table 8 shows how 2024 results change when one assumption is changed and the others are held at their base values. The figures come from re-running the model with each alternative value.

Table 8: Sensitivity of 2024 results ($ million)

Scenario	Sales	EBITDA	Profit after tax	Cash	Total assets
Base case	**19,184**	**4,335**	**2,688**	**2,930**	**17,800**
Sales growth	18,814	4,313	2,673	3,032	17,738

Scenario	Sales	EBITDA	Profit after tax	Cash	Total assets
2.33%					
Sales growth 4.33%	19,557	4,358	2,702	2,826	17,863
Cost slope 0.90	19,184	5,093	3,266	4,064	18,934
Cost slope 0.98	19,184	3,558	2,094	1,766	16,636
Tax rate 21.58% (lowest year)	19,184	4,335	2,781	3,115	17,986
Tax rate 26.15% (highest year)	19,184	4,335	2,618	2,791	17,661
Interest rate 3.5%	19,184	4,335	2,616	2,788	17,658

Source: model re-run with the stated input changed. Cost slope scenarios keep the regression intercept unchanged.

Profit is insensitive to sales growth. A one percentage point change in growth alters 2024 profit after tax by about $15 million, because each extra dollar of sales adds only six cents to EBITDA. Higher growth also reduces cash, since extra working capital and capital expenditure are needed to support the sales.

Profit is far more sensitive to the cost slope. Reducing the slope to 0.90 raises profit after tax to $3,266 million (21.5% higher) and increasing it to 0.98 reduces profit to $2,094 million (22.1% lower). The tax rate and the interest rate change profit by $70 to $95 million. Any user of the forecast should therefore concentrate on whether the cost relationship of 2018 to 2022 will persist. Colgate's cost of sales rose faster than sales in 2022, and if that was a temporary effect of input costs, the slope used here is too high.

Comparison with Reported 2023 Results and Limitations

Comparison with reported results

Colgate reported 2023 net sales of $19,457 million, growth of 8.3% on 2022, according to the

2023 results announcement noted on the Sales Growth Assumption sheet. The model assumed growth of 3.33% and forecast $18,565 million, 4.6% below the reported figure. Reported 2023 sales are above the model's 2024 forecast of $19,184 million. Applying the model's growth rate to the reported 2023 figure would give 2024 sales of about $20,105 million.

The shortfall arises because the growth assumption is based on the five-year record and a long-run industry forecast, and neither anticipates a year of growth well above trend. The method is designed for a stable growth path and does not capture one-off movements in prices or volumes. For forecasting beyond 2023, the base year should be updated to the reported 2023 statements.

Limitations

- **Cost structure.** The regression is fitted to five observations, and its slope of 0.94 implies almost no operating leverage. If the margin pressure of 2022 reverses, profit is understated, as Table 8 shows.
- **Cash accumulation.** Constant debt and equity mean that cash rises to $4.9 billion by 2027. In practice Colgate has returned cash through buybacks: net repurchases averaged about $0.8 billion a year over 2018 to 2022, and continuing at that rate would absorb most of the forecast cash build.
- **Base-year treatment of impairments.** The 2022 base excludes the $721 million impairment. This suits a forecast of recurring earnings, but it means the base year does not match the reported accounts.
- **Classification of debt.** The financing liabilities of $13,462 million include accruals, deferred taxes, other liabilities and noncontrolling interests. This overstates interest-bearing debt, so the model charges interest only on borrowings of $8,766 million. A different split between operating and financing liabilities would change the interest charge.
- **Noncontrolling interests.** Profit after tax is not split between Colgate shareholders and noncontrolling interests, and no distributions to minority holders are modelled. Net income attributable to noncontrolling interests was $182 million in 2022.

- **Currency and acquisitions.** Exchange rate effects and acquisitions are not forecast. Colgate's historical statements show both to be material in some years, for example $1,711 million spent on acquisitions in 2019.
- **Constant ratios.** Using five-year averages for working capital and fixed assets removes cyclical movement but also the 2022 build-up of inventories, which is reversed in 2023 in the model.

Conclusion

The pro forma statements for 2024 project net sales of $19,184 million, EBITDA of $4,335 million, profit after tax of $2,688 million, total assets of $17,800 million and operating cash flow of $3,220 million. The three statements are consistent with each other and the base year agrees with the published 2022 balance sheet.

The forecast rests on assumptions that are documented and supported by the 2018 to 2022 record, using the percent-of-sales method from the lecture. Its main weakness is the cost relationship: costs rose almost as fast as sales over the period, so the model produces flat profit regardless of sales growth. Comparison with reported 2023 sales shows that the sales assumption is cautious. A user who wanted a closer forecast should rebase the model on the 2023 reported statements and test the cost slope against the 2023 margin.

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Lecture materials, BBA2293 Financial Modelling: Building Financial Statements.

Appendix A: Workbook Map for Question 1

Sheet	Content
Income Statement, Balance Sheet, Cash Flow (Format 2)	Historical statements, 2018 to 2022 (inputs)
Sales Growth Assumption	Sales growth (industry forecast, CAGR) and interest rate on borrowings
Operating Cost Analysis	Operating costs as a percentage of sales; regression intercept and slope
NWC Analysis	Current operating assets and liabilities as a percentage of sales
Fixed Asset Analysis	Gross and net fixed assets as a percentage of sales
Depreciation Analysis	Average depreciation schedule (17 years)
Tax Rate Analysis	Annual, simple average and weighted average tax rates
Dividend Analysis	Dividend growth rate
Cash and Equivalents	Opening cash balance
Pro-forma	Assumptions (rows 4 to 18), income statement (rows 19 to 33) and balance sheet (rows 36 to 51)
Pro-forma Cash Flow	Pro forma cash flow statement and cash tie-out check

Appendix B: Key Formulas

Item	Formula
Sales	Prior-year sales x (1 + growth)
Operating costs excluding depreciation	-(intercept + slope x sales)
Depreciation	Prior-year gross fixed assets / depreciation schedule
Interest expense	Interest-bearing borrowings x interest rate
Interest earned	Prior-year cash x 1%

Item	Formula
Profit before tax	EBITDA - interest expense + interest earned - depreciation
Current assets and liabilities	Sales x ratio to sales
Net fixed assets	Sales x net fixed assets ratio
Gross fixed assets	Net fixed assets - accumulated depreciation (negative)
Retained earnings	Prior-year balance + profit after tax - dividends
Cash	Total liabilities and equity - all other assets
Capital expenditure	Increase in gross fixed assets